Homework 2

MATH 441: DIFFERENTIAL EQUATIONS

NAME: DATE:

(Exercises are taken from *Elementary Differential Equations and Boundary Value Problems* by William E. Boyce and Richard C. DiPrima)

1. Exercise §2.3 #4. A tank with a capacity of 500 gal originally contains 200 gal of water with 100 lb of salt in solution. Water containing 1 lb of salt per gallon is entering at a rate of 3 gal/min, and the mixture is allowed to flow out of the tank at a rate of 2 gal/min. Find the amount of salt in the tank at any time prior to the instant when the solution begins to overflow. Find the concentration (in pounds per gallon) of salt in the tank when it is on the point of overflowing. Compare this concentration with the theoretical limiting concentration if the tank had infinite capacity.

2. Exercise §2.4 #15. Solve the given initial value problem and determine how the interval in which the solution exists depends on the initial value y_0

$$y' + y^3 = 0, \qquad y(0) = y_0$$

3. Exercise §2.4 #28. The given equation is a Bernoulli equation. Solve it by using the substitution mentioned in Problem 27(b).

$$t^2y' + 2ty - y^3 = 0, \qquad t > 0$$

4. Exercise §2.5 #3. This problem involves equations of the form $dy/dt = f(y)$. Sketch the graph of $f(y)$ versus y, determine the critical (equilibrium) points, an classify each one as asymptotically stable or unstable. Draw the phase line, and sketch several graphs of solutions in the ty-plane

$$dy/dt = y(y-1)(y-2), \qquad y_0 \geq 0$$

The following problems involve equations of the form $dy/dt = f(y)$. Sketch the graph of $f(y)$ versus y, determine the critical (equilibrium) points, an classify each one as asymptotically stable, unstable, or semistable. Draw the phase line, and sketch several graphs of solutions in the ty-plane

5. Exercise §2.5 #10.

$$dy/dt = y(1 - y^2), \qquad -\infty < y < \infty$$

6. Exercise §2.5 #12.

$$dy/dt = y^2(4 - y^2), \qquad -\infty < y < \infty$$

7. Exercise §2.5 #20. At a given level of effort, it is reasonable to assume that the rate at which fish are caught depends on the population y: the more fish there are, the easier it is to catch them. Thus we assume that the rate at which fish are caught is given by Ey, where E is a positive constant, with units of 1/time, that measures the total effort made to harvest the given species of fish. To include this effect, the logistic equation is replaced by

$$dy/dt = r(1 - y/K)y - Ey$$

This equation is known as the **Schaefer model** after the biologist M.B. Schaefer, who applied it to fish populations.

(a) Show that if $E < r$, then there are two equilibrium points, $y_1 = 0$ and $y_2 = K(1 - E/r) > 0$.

(b) Show that $y = y_1$ is unstable and $y = y_2$ is asymptotically stable.

(c) A sustainable yield Y of the fishery is a rate at which fish can be caught indefinitely. It is a the product of the effort E and the asymptotically stable population y_2. Find Y as a function of the effort E; the graph of this function is known as the yield-effort curve.

(d) Determine E so as to maximize Y and thereby find the **maximum sustainable yield** Y_m.